

SETS/INDICES

i carrier, $i \in I$, where I is the set of carriers

j destination, $j \in J$, where J is the set of destinations

k price interval, $k \in K$, where K is the set of price intervals

t month, $t \in T$, where T is the set of months

PARAMETERS

P_{ijkt} = price per call minute from carrier i to destination j in price interval k , in month t

pen_{ijt} = penalty per call minute for carrier i to destination j in month t

d_{jt} = forecasted demand to destination j in month t

LT_{ik} = lower threshold of price interval for carrier i at interval k

UT_{ik} = upper threshold of price interval for carrier i at interval k

LB_{it} = Lower capacity bound for carrier i in month t

UB_{it} = Upper capacity bound for carrier i in month t

DECISION VARIABLES X_{ijkt} = # of call mins routed through carrier i to destination j in price interval k in month t

OBJECTIVE

$Y_{ik} = 1$ if carrier i operates in price interval k , otherwise 0

Minimize Cost

$$\sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} X_{ijkt} * (P_{ijkt} + \text{pen}_{ij,t})$$

CONSTRAINTS

Lower Capacity

$$\sum_{j \in J} \sum_{k \in K} X_{ijkt} \geq LB_{i,t}, \quad \forall i, t$$

Upper Capacity

$$\sum_{j \in J} \sum_{k \in K} X_{ijkt} \leq UB_{i,t}, \quad \forall i, t$$

Demand

$$\sum_{i \in I} \sum_{k \in K} X_{ijkt} \geq d_{j,t}, \quad \forall j, t$$

Lower Threshold

$$\sum_{j \in J} \sum_{t \in T} X_{ijkt} \geq LT_{ik} * Y_{ik}, \quad \forall i, k$$

Upper Threshold

$$\sum_{j \in J} \sum_{t \in T} X_{ijkt} \leq UT_{ik} * Y_{ik}, \quad \forall i, k$$

NON-NEGATIVITY

$$X_{ijkt} \geq 0$$

BINARY

$$Y_{ik} = \{0, 1\}$$

One Interval

$$\sum_{k \in K} Y_{ik} \leq 1$$